Parkinson's Disease Detection System Using Machine Learning Techniques

Harshit Mishra
Department of IT
KIET Group of Institutions Delhi NCR, India
harshitm312@gmail.com

Kundan Verma
Department of IT
KIET Group of Institution Delhi NCR, India
vermakundan156@gmail.com

Himanshu Yadav
Department of IT
KIET Group of Institution Delhi NCR, India
hy75790@gmail.com

Ajay Kumar
Department of IT
KIET Group of Institution Delhi NCR, India
ajaygarg100@gmail.com

***Abstract*—"Parkinson's disease is a chronic neurological disorder that has a major impact on an individual's motor functioning and has greatly impacted the quality of life due to debilitating symptoms, yet complex etiologies involving genetic and environmental elements. This paper examines the use of machine learning techniques for predicting Parkinson's disease in a voice dataset. We evaluated eight machine learning models, namely: Linear Regression, Logistic Regression, Decision Tree, Support Vector Machine (SVM), Random Forest, XGBoost, Neural Networks, and AdaBoost. The challenge of class imbalance has been addressed with the Self-Organizing Map Undersampling (SOMUS) technique. The models were evaluated using Recall, Precision, and F1 score. Our results indicate that the XGBoost and Random Forest classifiers performed best for parkinson's disease detection. The study indicates that machine learning can be applied for the improvement of diagnosis at an early stage and intervention in Parkinson's disease."**

I. INTRODUCTION

Parkinson's disease is a long-term degenerative neurological illness that mostly impairs motor control. A few main characteristics of the illness, such as slowed movement, stiffness in the muscles, and hand tremors, are what distinguish it most. These signs are brought on by the harm from this portion of the brain, the substantia nigra, which is a motor center that runs on dopamine. Parkinson's disease can worsen non-motor symptoms such as mood swings, impaired cognitive function, and altered autonomic nervous system in addition to its motor consequences. These symptoms can have a detrimental influence on the standard of living for those with a diagnosis with the condition [1]. Scientists believe that the fundamental cause is a complicated interaction between genetics and environment, despite the unknown processes that underlie it [2]. Disregarding that it is, in fact, the most widespread of all neurodegenerative ailments, up to now an uncurable medical malady, Parkinson's disease is. Various therapies such as drugs, surgical interventions, and supportive therapies are the ones mostly being applied in the therapy of many of its symptoms and improvement of patients' life quality as well.

Parkinson's disease, also known as affection of motion, is a neurological disorder that can cause a patient to have a variety of symptoms, including poor posture, trembling hands, muscular rigidity, hallucinations, and more. There is a common misperception that because sickness mostly affects older populations, early identification of illness cannot be implemented in younger populations. Medical statistics have battery increased manifold, the quantity supplied is from other clinical areas such as health care services. The need for these statistics is to handle them and to achieve insights from them through a Big Data analysis technique via Machine Learning approach that is supposed for handling various medical and clinical problems. Prior research has shown that popular models of machine learning algorithms, when given full treatment, perform better overall in categorization.

Furthermore, machine learning techniques have been applied in a number of prior research to predict various other types of disease such as diabetes prediction [3] and heart disease prediction [4]. However, the main subject of this study is to apply eight machine learning approaches for Parkinson's disease prediction.

The study's contributions are as follows:

- In this paper eight machine learning models were develope to predict Parkinson's disease.
- This paper assessed each model's performance using three performance metrics (Recall, Precision, and F1 score) across a dataset of Parkinson's disease cases.
- To address the issue of class imbalance in the Parkinson's disease data set, this study employs a recently proposed technique namely Self-Organized Map Under-Sampling(SOM-US) [5].

The organisation of the remaining portions of the research are include in Sections II to Section VII. The relevant work is presented in Section II, the proposed method, machine learning techniques, collection of data, and performance metric are explained in Section III, the experimental design is included in Section IV, Section V presented the results and discussion.

Finally, the research is concluded in Section VI with future work listed in Section VII.

II. RELATED WORK

Several notable studies have significantly advanced the area of Parkinson's disease diagnosis using machine learning.

- Mei et al. [6], states that the first comprehensive evaluation of all studies on the use of machine learning in the diagnosis of PD. It provides a high-level overview of the studies included: ML methods and their outcomes in diagnosis of Parkinson's Dieases, clinical behavioral and biometric data types that accurately pinpoint the disease, potential biomarkers for clinical decision-making, and relevant databases with which smaller collections may be expanded. The review will depict the immense potential of ML-driven PD diagnosis in advancing a systematic clinical decision-making process. Adoption of novel biomarkers can lead to earlier and more accessible PD diagnoses. General principles: ML approaches show a promising potential to provide clinicians with additional tools for screening, detection, and diagnosis of PD, potentially resulting in improved patient outcomes through earlier intervention and more accurate diagnoses.

- Aditi [7], states that the research examines the use of a Random Forest classifier to classify Parkinson's illness using vowel phonation data with a 91.835% accuracy and 0.95 sensitivity. Given that the 22 characteristics in the MDVP dataset have almost equal weights, the Random Forest model is the best choice. Similarly, the SVM model with PCA applied is able to achieve 91.836% accuracy and a sensitivity of 0.94. The top models all have strong resistance to outliers and zero false positive predictions. The KNN model also performs well given the balanced data. The study suggests the Random Forest model in classifying PD based on simplicity, accuracy, and non-invasive nature. Future enhancements will involve adding more data of audio and REM sleep data, as audio data alone is not sufficient for PD classification. These results promote the use of audio data generated on mobile devices for PD diagnosis via telemedicine with the aim of long-term patient relief.

- Alalayah et al. [8], states that Parkinson's disease, caused by a deficiency of dopamine, is hard to diagnose due to its unclear symptoms. Machine learning has helped with the early diagnosis of the disease from voice disorders. In that direction, the present study has made a contribution by providing a dataset including 22 voice characteristics, which were preprocessed to remove outliers and were further ranked using the Recursive Feature Elimination algorithm. Dimensionality reduction was done using t-SNE and PCA, and the resulting features were fed into DT, RF, MLP, KNN, and SVM classifiers. Better results were obtained, with RF combined with t-SNE getting 97% and MLP combined with PCA getting 98% accuracy. The findings demonstrate that sound waves have proven to effectively and automatically detect PD, likely to allow for early diagnosis and treatment to improve the lives of patients. This study adds to current knowledge with its new approach for early detection of PD through voice analysis.

- Shaban [9], states that this paper reviews the techniques of machine and deep learning applied to research in Parkinson's disease from 2016 to 2022, emphasizing neural networks due to their high performance and automated feature extraction. Key data modalities include sensory, handwriting, and EEG data. Most studies used binary classification (to differentiate PD patients from controls), and a few addressed PD staging or early biomarker identification. Early diagnostic attempts are hampered by a number of significant limitations, including limited sample sizes, clinician diagnosis heterogeneity, and a preponderance of individuals with severe Parkinson's disease in the datasets. Future studies should be performed on clinically relevant datasets, including DaTscan, and should try other modalities, such as sleep EEG, to help in the early biomarker identification and better management of PD. In order to obtain early diagnosis and therapy of Parkinson's disease (PD), this article highlights the need for larger and more diversified datasets as well as more sophisticated approaches.

- Chatterjee et al. [10], states that early detection of Parkinson's disease is of prime importance for understanding its etiology and enabling early treatment. In this work, an ensemble-based PDD-ET model, using premotor features, is proposed to discriminate between healthy individuals and those with PD. The PDD-ET model has reached high accuracy of 95.325%, far better than those of 14 different machine learning and deep learning models. This represents a superior performance credited to the conglomeration of various machine learning and deep learning techniques, which enhanced its ability for PD diagnosis. By delivering medicines to patients at an early stage of the illness, the experimental results demonstrate the PDD-ET model's efficacy in early PD diagnosis and highlight its potential to improve patient outcomes.

- Latifoğlu et al. [11], states that this study focuses on the application of the Variational Mode Decomposition technique to EEG signals in the automatic diagnosis of Parkinson's disease. EEG signals from PD patients, both ON and OFF medication, and healthy controls were decomposed into sub-bands; various features were extracted, including SE, HPs, PS, KC, and KFD. Chi-squared tests were applied to identify significant features, and input into four different machine learning algorithms

to classify PD ON-HC and PD OFF-HC groups was done for the analysis of the effects of dopaminergic drugs. Comparison between this VKF method and the standard methods, namely EMD and VMD, proved that it outperformed them all over the ML algorithms applied. This new application of the VKF method for the diagnosis of neurological disorders using EEG signals reveals its potential. Future studies will involve large cohorts of subjects and clinician consultation for validation and extension of these results.

III. RESEARCH METHODS

A. Proposed Method

This study proposes a model for Parkinson's disease prediction based on machine learning techniques. For the validation of the proposed model, this study make use of eight machine learning approaches on a Parkinson's disease data set. For the evaluation of these eight parkinson's disease prediction models, three performance measures (F1 score, recall, and precision) were used. Figure 1 depicts the general overview of the suggested approach.

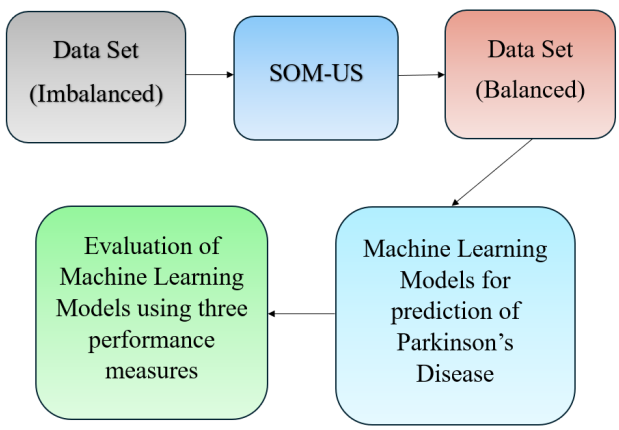

Fig. 1. Proposed Methodology

B. Machine Learning Techniques

In our research we used eight different machine learning techniques to detect parkinson's disease. These techniques are Linear Regression, Logistic Regression, Decision Tree, Support Vector Machine (SVM), Random Forest, XGBoost, Neural Networks, and AdaBoost. With regard to handling intricate medical data, each approach offers unique advantages that enhance the prediction model's overall precision and dependability.

C. Data Set

1) Brief Description: A substantial amount of biological speech data from thirty-one subjects—23 of whom have Parkinson's disease—has been collected. The study's data set was obtained from kaggle [12].There are 195 voice recordings in the dataset; each recording has a row in the table matching to the person's name and columns containing different voice metrics. Using this data, our main objective is to differentiate between those who have Parkinson's disease and those who are healthy, as indicated by the "status" column, which indicates if the individual has Parkinson's Disease (1) or is healthy (0).

2) Pre-processing: One of the biggest difficulties we encountered was handling class imbalance,This is a frequent problem in medical datasets since there are typically more healthy samples than sick ones. To address this, we used a recent technique called SOM-US [5].This method balance the dataset by reducing the dominance of the majority class preserving the minority class's integrity, which enhanced the model's capacity to recognise Parkinson's disease.

D. Performance Measures

This study evaluates eight machine learning methods for Parkinson's disease prediction using three performance metrics: recall, precision, and F1 score.
Details of these performance measures are below:

Recall indicates the model's ability to identify actual cases of Parkinson's disease, which is important to ensure that no one who need assistance is overlooked.

$$\text{Recall} = \frac{\text{True Positive(TP)}}{\text{True Positive(TP)} + \text{False Negative(FN)}}$$

Precision provides an indication of the model's predictive accuracy by indicating the proportion of correctly predicted cases.

$$\text{Precision} = \frac{\text{True Positive(TP)}}{\text{True Positive(TP)} + \text{False Positive(FP)}}$$

F1-score provides us with a single figure that balances recall and accuracy, enabling us to view the model's total performance.

$$\text{F1 Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

IV. EXPERIMENTAL DESIGN

In order to verify the suggested methodology, an experimental study has been conducted using eight machine learning techniques, three performance measures, and one Parkinson's Disease data set. An overview of the experimental design shown in Fig. 2.

The issue of unbalanced data in Parkinson's disease detection is also taken into account in the experimental design. We used the SOM-US technique to pre-process the data in order to obtain a balanced number of both classes(majority and minority).Given that the dataset has an unequal distribution of positive and negative examples,

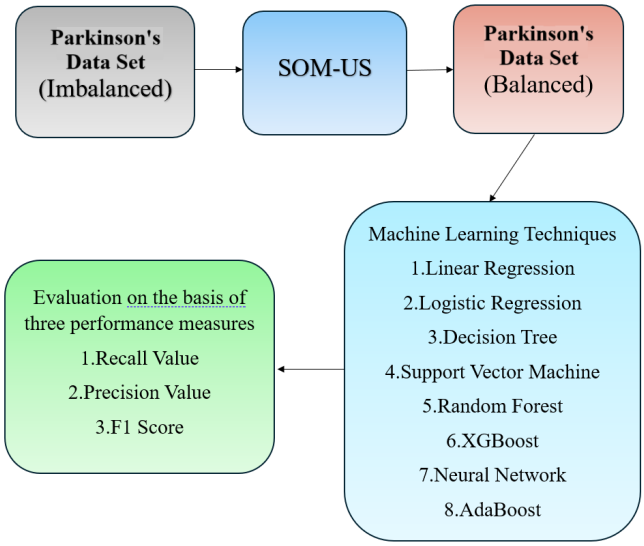

Fig. 2. Experimental Design

taking this step is crucial to reducing bias. These processed data were then thoroughly tested with several machine learning algorithms to determine the most accurate model for Parkinson's disease detection. Based on three performance metrics — (Recall, Precision, and F1 score) we determined the top model.

V. Results & Discussion

Table I displays the outcomes of eight machine learning models for parkinson's disease prediction, with respect to three performance measures.

TABLE I
Outcomes of eight Machine Learning Models

S.No	Algorithms	Recall	Precision	F1-score
1	Linear Regression	0.55	0.30	0.39
2	Logistic Regression	0.02	0.20	0.05
3	Decision Tree	0.50	0.48	0.49
4	Support Vector Machine	0.02	0.20	0.05
5	Random Forest	0.50	0.89	0.64
6	XGBoost	0.52	0.81	0.64
7	Neural Network	0.41	0.77	0.52
8	AdaBoost	0.49	0.80	0.62

From Table I following observations are drawn.

- ""The models provided relevant information and performed exceptionally well in diagnosing Parkinson's sickness. The performance of XGBoost is slightly better than the Random Forest classifier. The XGBoost model has a high F1 score of 0.64, indicating a good balance between precision and recall".
- "Recall and the F1 score demonstrate that XGBoost performs better than Random Forest. However, it has a slightly lower precision than Random Forest; that is, it is better at capturing true positives, although less precise at the same time. The Random Forest was better in precision but suffered in terms of recall".

From the above observations it can be concluded that the XGBoost and Random Forest classifiers provided the best performance.

VI. Conclusion

This study developed the eight machine learning models — Linear Regression, Logistic Regression, Decision Tree, Support Vector Machine (SVM), Random Forest, XGBoost, Neural Networks, and AdaBoost for parkinson's disease detection. One of the critical problems related to the studied dataset was its class imbalance. This issue was dealt with by applying the Self-Organizing Map Undersampling (SOM-US) technique.

For the evaluation of eight machine learning models, three performance measures (Recall, Precision, and F1 score) were used. Our experimental results shows that XGBoost and Random Forest performed the best for parkinsin's disease prediction.

VII. Future Works

As a future work, this study can be extended towards the integration of other data types, such as genetic and imaging data, for the improvement of model accuracy and robustness. Further, increasing the size of the dataset and integrating longitudinal data will help to elucidate the nature of disease progression and improve early diagnosis capabilities. Continued development in machine learning and data processing techniques will be central to the development of better diagnostic tools for Parkinson's disease and other intractable medical conditions.
Additionally, using the large number of data sets may be the matter of future work extended towards this study.

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